

Chapter 1

Definition

Let $C(X)$ be functions from a metric space (X, d) to $\mathbb{R}$ or $\mathbb{C}$.

Definition 1.1 (Bounded Class of Functions). Let (X, d) be a metric space. $\mathcal{F} \subset C(X)$ is bounded if and only if $\exists M > 0$ such that $\forall f \in \mathcal{F}, \|f\|_\infty < M$

Definition 1.2 (Equicontinuous Class of Functions). Let (X, d) be a metric space. $\mathcal{F} \subset C(X)$ is equicontinuous if and only if $\forall \varepsilon > 0$ there exists a $\delta > 0$ such that $\forall f \in \mathcal{F}$ and $\forall x, y \in X$ such that $d(x, y) < \delta$ then $|f(x) - f(y)| < \varepsilon$.

Chapter 2

Main Theorem

Theorem 2.1 (Main Theorem). Let (X, d) be a compact metric space. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of bounded, equicontinuous functions then there exists a subsequence which is uniformly convergent ie convergent with respect to $|\cdot|_\infty$.

Proof: By Lemma 2.2, there exists a subsequence $(g_n)_{n=1}^\infty$ which converges pointwise on a countable dense subset A of X . By Lemma 2.3, $(g_n)_{n=0}^\infty$ is uniformly convergent. $\square$

Lemma 2.2 (Pointwise Convergent Subsequence on Countable Dense Subset). Let (X, d) be a compact metric space. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence which is bounded then there exists a subsequence which is pointwise convergent on a countable dense subset of X .

Proof: Since X is compact, it is totally bounded which implies it is separable. You create a countable dense subset by taking the union of the centers of the finite set of balls with radii $\frac{1}{n}$. Let $A = \{a_i\}_{i \in \mathbb{N}}$ be the countable dense subset. Since $(f_n)_{n \in \mathbb{N}}$ is bounded, there exists a $M > 0$ such that $\|f_n\|_\infty < M$ for all n .

Diagonalization Argument: We will be defining a series of subsequences such that the n th subsequence has pointwise convergent on $\{a_1, \dots, a_n\}$. Let $f_n^{(0)} = f_n$. By boundedness of $f_n^{(0)}$, $(f_n^{(0)}(a_1))_{n \in \mathbb{N}}$ is a sequence in $\overline{B}_M(0)$, which is closed and bounded hence compact (Heine Borel), thus there exists a subsequence $(f_n^{(1)})_{n \in \mathbb{N}} \subset (f_n^{(0)})_{n \in \mathbb{N}}$ which is such that $(f_n^{(1)}(a_1))_{n \in \mathbb{N}}$ is convergent in $\overline{B}_M(0)$. Assume that $(f_n^{(m)})_{n \in \mathbb{N}}$ is a subsequence of $(f_n)_{n \in \mathbb{N}}$ such that $(f_n^{(m)}(a_i))_{n \in \mathbb{N}}$ converges for $i \in \{1, \dots, m\}$. $(f_n^{(m)}(a_{m+1}))_{n \in \mathbb{N}}$ is a sequence in $\overline{B}_M(0)$, hence there exists a subsequence $(f_n^{(m+1)})_{n \in \mathbb{N}}$ such that $(f_n^{(m+1)}(a_{m+1}))_{n \in \mathbb{N}}$ converges.

Thus there exists a sequence of subsequences $(f_n^{(0)}) \supseteq (f_n^{(1)}) \supseteq \dots$, such that $(f_n^{(m)})_{n \in \mathbb{N}}$ converges pointwise on $\{a_1, \dots, a_m\}$. Construct the subsequence (g_n) of (f_n) , where $g_n = f_n^{(n)}$. Note that $\forall k \geq n$, g_k is a subsequence of $f_m^{(n)}$ and thus converges pointwise on $\{a_1, \dots, a_n\}$. Hence g_k converges pointwise on all of A . $\square$

Lemma 2.3 (Equicontinuous sequence + Pointwise $\Rightarrow$ Uniform Convergence).

Let (X, d) be a compact metric space. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence which is equicontinuous and converges pointwise on a countable dense subset $A = \{a_1, a_2, \dots\}$. $(f_n)_{n \in \mathbb{N}}$ converges uniformly with respect to the $\|\cdot\|_\infty$.

Proof: $\forall \varepsilon > 0$, by equicontinuity there exists a $\delta > 0$ such that $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \frac{\varepsilon}{6}$ for all $n \in \mathbb{N}$. $\forall x \in X$, by density of A : $\exists a \in A \cap B_\delta(x)$. Thus $x \in B_\delta(a)$. Thus

$$X = \bigcup_{i=1}^{\infty} B_\delta(a_i).$$

By compactness, there exists a finite subcover of balls with radii δ with centers $\{c_1, \dots, c_k\} \subset A$. By pointwise convergence and hence Cauchy convergence at c_i , $\exists N_i$ such that $\forall m \geq n \geq N_i$:

$$|f_n(c_i) - f_m(c_i)| < \frac{\varepsilon}{6}.$$

Take $N = \max\{N_1, \dots, N_k\}$.

$\forall x \in X$, there exists a c_i such that $d(x, c_i) < \delta$. Furthermore, $\forall m \geq n \geq N$ we have the following:

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq |f_n(x) - f_n(a_i)| + |f_n(a_i) - f_m(a_i)| + |f_m(a_i) - f_m(x)| \\ &< \frac{\varepsilon}{6} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} < \frac{\varepsilon}{2} \Rightarrow \|f_n - f_m\|_\infty \leq \frac{\varepsilon}{2} < \varepsilon \end{aligned}$$

Hence $(f_n)_{n=0}^\infty$ is uniformly cauchy hence uniformly convergent.

□

Theorem 2.4. Let (X, d) be a compact metric space. $\mathcal{F} \subset C(X)$ is compact if and only if $\mathcal{F}$ is equicontinuous, bounded, and closed

Proof: $\Rightarrow$: Suppose $\mathcal{F}$ is compact. The function $\|\cdot\|_\infty : C(X) \rightarrow [0, \infty)$ is continuous, hence $\exists k \in \mathcal{F}$ such that $\sup_{f \in \mathcal{F}} \|f\|_\infty = \|k\|_\infty$ thus $\mathcal{F}$ is bounded. By total boundedness $\forall \varepsilon > 0$, there exists functions $f_1, \dots, f_n \in \mathcal{F}$ such that

$$\mathcal{F} = \bigcup_{i=1}^n B_{\frac{\varepsilon}{3}}(f_i).$$

Since f_i are continuous on X a compact space, they are uniformly continuous. Thus $\exists \delta_i > 0$ such that $\forall x, y \in X$ where $d(x, y) < \delta_i$: $|f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$. Let $\delta = \min\{\delta_1, \dots, \delta_n\}$. $\forall g \in \mathcal{F}$, there exists a f_i such that $\|g - f_i\|_\infty < \frac{\varepsilon}{3}$. $\forall x, y \in X$ where $d(x, y) < \varepsilon$:

$$\begin{aligned} |g(x) - g(y)| &\leq |g(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - g(y)| \\ &\leq 2\|g - f_i\| + |f_i(x) - f_i(y)| \\ &< 2\left(\frac{\varepsilon}{3}\right) + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

$\Leftarrow$: Apply Theorem 2.1 (Arzela-Ascoli theorem) and then sequential compactness.

(Alternative Proof): Suppose $\mathcal{F}$ is equicontinuous, bounded, and closed. $\mathcal{F}$ is complete. So to show $\mathcal{F}$ is compact we must show that it is totally bounded.

$\forall \varepsilon > 0$. By equicontinuity, there exists $\delta > 0$ such that if $d(x, y) < \delta$: $|f(x) - f(y)| < \frac{\varepsilon}{6}$ for all $f \in \mathcal{F}$. Since X is compact we can cover it with δ balls centered at $x_1, \dots, x_k \in X$. Since $\mathcal{F}$ is bounded, there exists a $M > 0$ such that $\forall f \in \mathcal{F}$: $f(X) \subseteq \overline{B}_M(0)$. Since $\overline{B}_M(0)$ is compact in $\mathbb{K}$, there exists a covering of $\frac{\varepsilon}{12}$ balls centered at $y_1, \dots, y_l$. Define the function $\varphi_f : \{1, \dots, k\} \rightarrow \text{Pow}(\{1, \dots, l\})$ where $\varphi_f(i) = \{j \in \{1, \dots, l\} : f(x_i) \in B_{\frac{\varepsilon}{12}}(y_j)\}$. Note that there are only finite number of such functions. You can say that $\varphi : \mathcal{F} \rightarrow \text{Hom}(\{1, \dots, k\}, \text{Pow}(\{1, \dots, l\}))$ where $f \rightarrow \varphi_f$ defines an equivalence relation. Suppose $f, g \in \mathcal{F}$ where $\varphi_f = \varphi_g$. $\forall x \in X$, there exists $i \in \{1, \dots, k\}$ such that $d(x, x_i) < \delta$. Furthermore since $\varphi_f(x_i) = \varphi_g(x_i) \neq \emptyset$, then $\exists y_j \in \varphi_f(x_i) = \varphi_g(x_i)$. Thus:

$$\begin{aligned}
 |f(x) - g(x)| &= |f(x) - f(x_i) + f(x_i) - g(x_i) - g(x)| \\
 &\leq |f(x) - f(x_i)| + |f(x_i) - g(x_i)| + |g(x_i) - g(x)| \\
 &\leq |f(x) - f(x_i)| + |f(x_i) - y_j| + |y_j - g(x_i)| + |g(x_i) - g(x)| \\
 &< \frac{\varepsilon}{6} + \frac{\varepsilon}{12} + \frac{\varepsilon}{12} + \frac{\varepsilon}{6} \\
 &= \frac{\varepsilon}{2} < \varepsilon
 \end{aligned}$$

Thus $\|f - g\|_\infty < \varepsilon$. There are finite number of equivalence classes, thus take one representative of each equivalence class as your centers of your ε balls.

□

Chapter 3

Examples

Chapter 3

References